# $C^2F$-VRWKV: A Lightweight Clustering-Aware and Cross-Frequency-Enhanced Vision-RWKV Network for Ship Detection in Remote Sensing Scenarios

Shounan Wang, Chen Jia, *Member, IEEE*, Hui Liu, *Graduate Student Member, IEEE*, Xu Cheng, *Senior Member, IEEE*, Fan Shi, *Member, IEEE*, and Shengyong Chen, *Senior Member, IEEE*

***Abstract*—Accurate ship detection in remote sensing imagery is crucial for maritime traffic monitoring, but is hindered by complex backgrounds and the small, variable size of ships. Existing methods often struggle to balance detection accuracy with computational efficiency. To address these limitations, we propose a novel lightweight vision-receptance weighted key value (VRWKV) network with cross-frequency and clustering-aware enhancements ($C^2F$-VRWKV) for efficient ship detection. Specifically, $C^2F$-VRWKV integrates three core components: first, a $C^2$-VRWKV backbone incorporates a contextual clustering mechanism to effectively aggregate local fine-grained morphological features within a linear-complexity scanning paradigm. Second, the cross-frequency fusion module (CFFM) employs a frequency-guided cross-attention strategy to decouple and emphasize ship contours and textures while effectively suppressing complex environmental noise. Third, the dual-stream multifrequency attention (DS-MFA) module utilizes a prior-guided mechanism to separate foreground ship features from background maritime clutter, enabling robust modeling across diverse scales. Experiments on the Airbus-Ship dataset demonstrate that $C^2F$-VRWKV achieves 78.23% mAP with just 0.53M parameters, outperforming SOTA methods and offering a resource-efficient solution for ship detection in constrained settings. This work offers an effective and lightweight solution to the challenging task of ship detection in complex maritime environments.**

***Index Terms*—Contextual clustering, lightweight model, remote sensing images, small ship detection, vision-receptance weighted key value (VRWKV).**

## I. Introduction

ACCURATE ship detection in optical remote sensing imagery is crucial for efficient maritime traffic monitoring and fishery resource management [1], [2]. However, the complex and dynamic nature of maritime environments leads to significant variations in the scale and visual appearance of ship targets. Consequently, achieving precise ship localization in remote sensing images remains a formidable challenge [2], [3]. Recently, convolutional neural networks (CNNs), such as VGG [4], Faster R-CNN [5], and YOLO [6], have demonstrated remarkable capability in feature extraction and object recognition, opening new technical avenues for ship detection in remote sensing imagery. Specifically, shallow CNN layers retain high spatial resolution and smaller receptive fields, enabling accurate localization of small-scale ships in optical remote sensing images [7], [8]. However, the inherent locality of conventional CNNs restricts their ability to capture global contextual information in complex scenes [9], [10], [11], thereby reducing their discriminative power against maritime clutter and often leading to missed detections of ship targets. The success of the Vision Transformer (ViT) [12], [13], [14] has demonstrated the superior capability of Transformers in modeling long-range contextual dependencies, which is crucial for achieving precise ship localization under vast and complex maritime backgrounds [9]. However, as the sequence length increases, their computational and memory costs grow quadratically, leading to significant resource consumption and computational bottlenecks when processing high-resolution remote sensing imagery, thereby severely constraining their deployment on resource-limited onboard satellite platforms [15], [16]. Although several efficient Transformer variants, such as Sparse Transformer [17] and Linear Transformer [18], attempt to linearize attention through approximation or low-rank decomposition to reduce computational cost, these simplifications often come at the expense of modeling fine-grained spatial details and weak object features, thereby hindering accurate detection of small-scale ships. Consequently, existing Transformer-based approaches still struggle to achieve an optimal balance between detection accuracy and computational efficiency.

Recently, linear-attention models represented by receptance weighted key value (RWKV) [19] have attracted increasing attention for their strong sequence modeling capability under low computational complexity. Inspired by this paradigm, vision-RWKV (VRWKV) [20] successfully extends RWKV to the visual domain. However, in remote sensing imagery, ship targets exhibit highly localized structures and arbitrary spatial distributions, posing significant challenges to the core block design and 1-D scanning mechanism of existing visual RWKV

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The authors are with the School of Computer Science and Engineering, Tianjin University of Technology, Tianjin 300386, China (e-mail: jiachen@email.tjut.edu.cn; xu.cheng@ieee.org).

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models. Most RWKV-based approaches rely on global linear scans that flatten 2-D feature maps into 1-D sequences, limiting the model's ability to selectively aggregate local contextual information. This limitation is particularly detrimental to small ships, as insufficient local context aggregation weakens the model's discrimination between fine-scale targets and background noise. Moreover, common row-wise or four-directional scanning strategies fail to preserve semantic proximity when handling ships with nonaligned and multidirectional spatial structures, reducing robustness under multiscale and cluttered maritime scenes. Consequently, existing visual RWKV models still suffer from false detections and localization errors, especially for small-scale ship targets.

To address these challenges, we propose $C^2F$-VRWKV, a lightweight network capable of achieving high-precision ship detection in remote sensing imagery with minimal computational cost. Specifically, we design a novel $C^2$-VRWKV backbone that introduces a contextual clustering mechanism to adaptively aggregate local semantics within windowed regions, enhancing the discrimination of fine ship structures while preserving the long-range dependency modeling of RWKV. To address challenges from complex maritime backgrounds, we develop a cross-frequency fusion module (CFFM) that employs a frequency-guided cross-attention strategy to decouple and emphasize high-frequency components representing ship contours and textures, effectively suppressing clutter from waves and clouds. Furthermore, a dual-stream multifrequency attention (DS-MFA) module is introduced, which incorporates a foreground–background separation strategy to jointly model multiscale ships with fine-grained attention on the foreground while suppressing background interference. These designs enable $C^2F$-VRWKV to achieve robust and efficient multiscale ship detection under complex maritime conditions.

The primary contributions of our work are summarized as follows.

1) We propose $C^2F$-VRWKV, a lightweight VRWKV network that accurately localizes ship targets in complex maritime backgrounds while maintaining a low parameter count and computational overhead.
2) We design a $C^2$-VRWKV backbone to enhance small ship discrimination through local–global aggregation; a CFFM to suppress complex backgrounds by emphasizing high-frequency details; and a DS-MFA to enable robust multiscale modeling via foreground–background separation, ensuring high accuracy and efficiency in challenging conditions.
3) As shown in Fig. 1, we evaluate $C^2F$-VRWKV on benchmark remote sensing datasets across diverse maritime scenarios, with results demonstrating that our method outperforms other SOTA methods while maintaining a remarkably low parameter count.

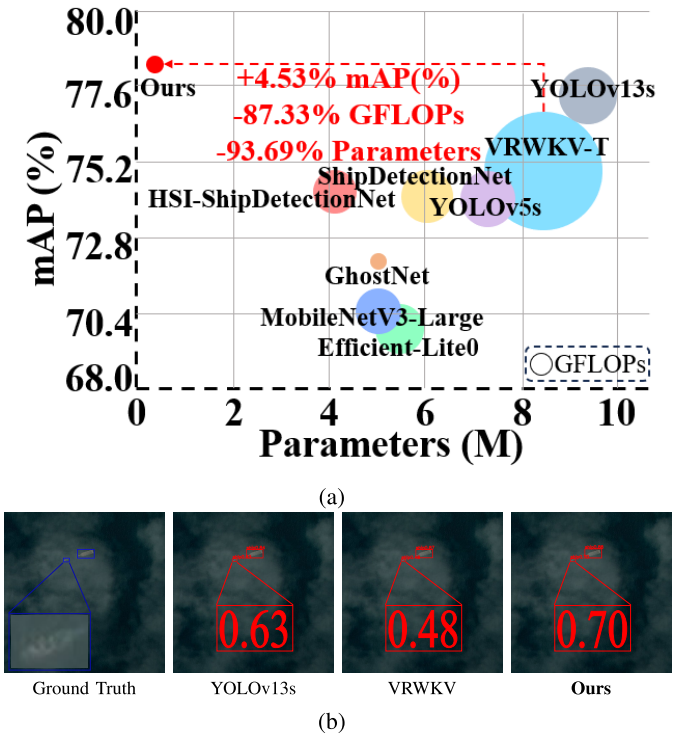



Fig. 1. Performance of $C^2F$-VRWKV on the Airbus-Ship dataset. (a) Comparison with SOTA methods. (b) Comparison of confidence scores under complex scenarios.

## II. Related Works

### A. Ship Detection Network

With the rapid development of deep learning, CNN-based detectors have become the dominant paradigm in ship detection, achieving SOTA performance [21], [22], [23], [24]. For instance, Changhua et al. [25] enhanced small-object recognition by improving the residual connections and FPN structure of YOLOv3. To address the multiscale challenge, Tian et al. [26] introduced a GAN-based image enhancement unit and an expanded receptive field strategy to improve feature extraction across varying ship sizes. Similarly, Xie et al. [27] proposed an adaptive feature enhancement module to strengthen the localization ability of deep features. Despite their strong representation power, CNN-based methods inherently rely on local operations with limited receptive fields, making it difficult to capture long-range contextual cues and effectively suppress large-scale maritime clutter [28].

In contrast, Transformer architectures [13] have shown great potential for modeling long-range dependencies through self-attention and have recently gained popularity in ship detection networks [29], [30], [31]. For example, Wang et al. [32] integrated a feature transformation module (FTM) into YOLOX to enhance contextual modeling. However, the quadratic computational complexity of the self-attention mechanism with respect to the number of image tokens [33] leads to substantial parameter counts and high computational costs, thereby limiting the deployment of Transformer-based models on resource-constrained edge platforms that require real-time processing.

### B. Vision RWKV Models

To bridge the gap between the locality limitation of CNNs and the computational bottleneck of Transformers, a new architecture called RWKV [19] has emerged, combining the

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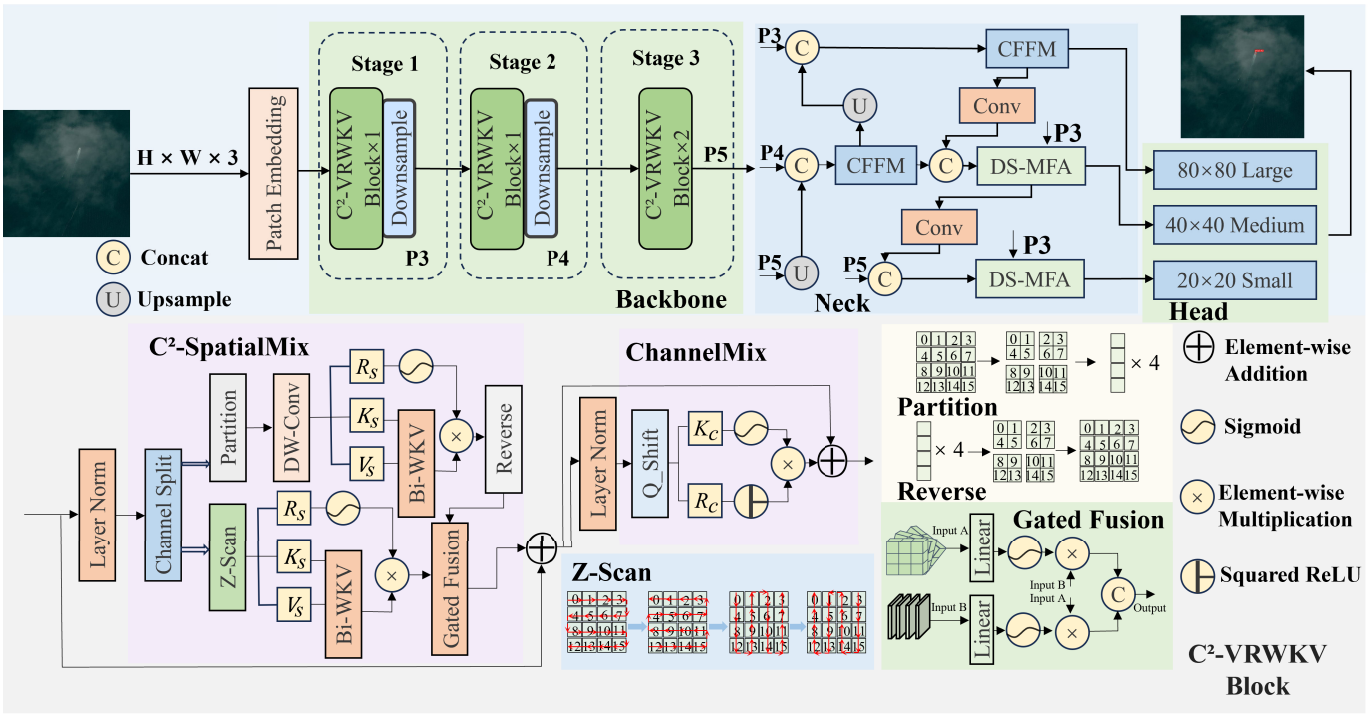



Fig. 2. Overview of the proposed $C^2F$-VRWKV architecture. The input image is first processed by the lightweight backbone $C^2$-VRWKV, which incorporates a contextual clustering mechanism to aggregate local and global semantic features, thereby enhancing the representation of small ships. The extracted features are then intelligently fused by a cascaded neck network, which integrates the CFFM to suppress complex backgrounds and the DS-MFA module to model ships across multiple scales, generating refined multiscale feature maps. These feature maps are subsequently fed into a YOLO-based detection head, which performs final classification and bounding-box regression. The head operates on these enhanced feature maps to produce multiscale predictions, which are then processed through nonmaximum suppression to yield the final detection results for ships of various sizes.

efficiency of RNNs with the expressive power of Transformers. Equipped with a linear-complexity attention mechanism and parallelizable training strategy, RWKV has achieved remarkable success in natural language processing (NLP) tasks [34]. Recently, researchers have extended RWKV to various vision domains, demonstrating strong potential. For instance, diffusion-RWKV [35] integrates RWKV with diffusion models for improved computational efficiency, Restore-RWKV [36] applies it to medical image restoration, and RWKV-CLIP [37] achieves impressive performance on multimodal retrieval tasks. In computer vision, VRWKV [20] adapts this paradigm into a general-purpose backbone through a bidirectional global attention mechanism (Bi-WKV) and a quad-directional token shift (Q-Shift). However, when the model is directly applied to ship detection in remote sensing imagery, where fine-grained spatial structures are crucial for precise localization, its strong bias toward global modeling reveals a critical limitation in capturing local contextual information. Specifically, existing VRWKV models rely on global linear scanning across feature maps, which flattens 2-D spatial structures into 1-D sequences. While this enables effective long-range dependency modeling, it inevitably weakens local neighborhood preservation. Consequently, the model struggles to discriminate small or low-contrast ships from complex maritime backgrounds, as it lacks a dedicated mechanism for aggregating localized contextual cues such as hull contours or edges. This often leads to false positives and missed detections. Therefore, although VRWKV provides a promising direction for efficient visual modeling, reintegrating localized context modeling into its core architecture is crucial for realizing its full potential in complex detection scenarios.

## III. Methodology

### A. Preliminary

The complete architecture of our proposed $C^2F$-VRWKV is depicted in Fig. 2. It includes two main components: 1) the $C^2$-VRWKV backbone for aggregating robust ship features and 2) the cascaded neck network for efficient multiscale feature fusion. To address complex backgrounds and multiscale challenges, we integrate the CFFM in the top-down fusion path and the DS-MFA in the bottom-up refinement path.

For a single image $I \in \mathbb{R}^{H\times W\times 3}$, spatial information is transformed via patch embedding, forming a sequence of tokens $T_{\text{in}} = \{t_1, t_2, \ldots, t_N\}$. This sequence is processed through the $C^2$-VRWKV backbone, embedding key ship cues from both local and global contexts into multiscale feature maps $\{P3, P4, P5\}$. Finally, in the cascaded neck, these feature maps are intelligently fused and refined, producing high-precision bounding box predictions $B_{\text{out}} = \{b_1, b_2, \ldots, b_k\}$.

### B. $C^2$-VRWKV Block

The proposed $C^2$-VRWKV block serves as the core building block of our $C^2$-VRWKV backbone, designed to structurally address the deficiencies of the original VRWKV in local

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context modeling. As shown in Fig. 2, it adopts a typical Pre-LN residual structure, and its computational process can be formulated as

$$X' = X + DropPath\left(C^2\text{-}SpatialMix\left(LN\left(X\right)\right)\right) \quad (1)$$

$$X_{\text{out}} = X' + DropPath\left(ChannelMix\left(LN\left(X'\right)\right)\right) \quad (2)$$

where $X \in \mathbb{R}^{B\times N\times C}$ represents the input feature sequence, with $B$ denoting the batch size, $N$ signifying the sequence length, and $C$ being the channel dimension. The operator LN(·) stands for layer normalization, while DropPath(·) refers to the stochastic depth regularization operation.

In this design, ChannelMix(·) follows the standard RWKV design for channel-wise interaction, while our core innovation is concentrated in the context-aware spatial mixing module, $C^2$-SpatialMix(·). This module introduces a hybrid attention mechanism with local–global decoupling, which effectively fuses local neighborhood information with long-range dependency features. Given an input feature $Z \in \mathbb{R}^{B\times N\times C}$, $C^2$-SpatialMix first splits it along the channel dimension into two complementary subspaces: 1) $Z_{\text{l}} \in \mathbb{R}^{B\times N\times C_{\text{l}}}$ for local contextual clustering and 2) $Z_{\text{g}} \in \mathbb{R}^{B\times N\times C_{\text{g}}}$ for global dependency scanning. Rather than employing a static channel division across all layers, our $C^2$-VRWKV backbone utilizes a stage-adaptive progressive allocation strategy to dynamically balance local and global representational capacities as the network deepens. This specific design choice is inherently guided by the hierarchical nature of feature abstraction in remote sensing imagery. In the shallow layer (Stage 1), extracting fine-grained morphological details is equally as important as preliminary background modeling. Therefore, we adopt a symmetrical 1:1 division ($C_l = C_g = 0.5C$). As the network progresses to the intermediate layer (Stage 2) and the spatial resolution decreases, the primary objective shifts toward building broader semantic dependencies. To reflect this, we reduce the local branch allocation to 25% ($C_l = 0.25C$) and expand the global scanning branch to 75% ($C_g = 0.75C$). Finally, in the deepest layer (Stage 3), where spatial dimensions are highly compressed, applying localized window operations becomes structurally redundant. Hence, we entirely disable the local clustering branch ($C_l = 0$) and allocate 100% of the channels to global semantic abstraction. This progressive transition from focusing on microdetails in shallow stages to capturing pure macrosemantics in deep stages ensures an optimal utilization of channel dimensions and computational efficiency. After the channel split, the local branch reshapes $Z_{\text{l}}$ into a 2-D feature map and partitions it into multiple nonoverlapping windows, each of size $M \times M$. A lightweight depth-wise separable convolution (DWConv) is then applied within each window to extract local representations

$$\mathcal{L} = DWConv\left(window_partition\left(Z_{\text{l}}, M\right)\right). \quad (3)$$

This process operates directly in 2-D space, preserving the topological relationships between pixels and thereby enabling the efficient capture of fine-grained morphological features such as hull contours and edges. Concurrently, the global branch inherits the advantages of RWKV. It maintains spatial proximity by applying Zigzag Scanning to $Z_{\text{g}}$ and subsequently generates the receptance, key, and value vectors through linear mappings

$$R_{\text{g}}, K_{\text{g}}, V_{\text{g}} = f_R\left(Z_{\text{g}}\right), f_K\left(Z_{\text{g}}\right), f_V\left(Z_{\text{g}}\right). \quad (4)$$

Subsequently, the global contextual representation is computed with linear complexity using the efficient Bi-WKV operator. The computation at each timestep $t$ is formulated as

$$\mathcal{G}_t = \sigma\left(R_{\text{g},t}\right) \odot \frac{\sum_{i=1}^{N} \exp\left(w_{t,i}\right) K_i V_i}{\sum_{i=1}^{N} \exp\left(w_{t,i}\right) K_i} \quad (5)$$

where $\mathcal{G}_t \in \mathbb{R}^{C_{\text{g}}}$ represents the global contextual output for the $t$th token; $\sigma(\cdot)$ is the Sigmoid function for gating modulation; $\odot$ denotes element-wise multiplication; and $w_{t,i}$ is the learnable time-decay weight term, which is jointly determined by the relative position and learnable parameters, enabling the model to capture long-range dependencies with linear complexity.

To achieve adaptive fusion of local and global features, we design a bidirectional gated fusion mechanism. The local representation $\mathcal{L}$ and the global representation $\mathcal{G}$ mutually generate gating signals to modulate each other's information flow. This fusion process is defined as

$$\hat{\mathcal{L}} = \mathcal{L} \odot \sigma\left(\mathcal{G} W_{g\to l}\right) \quad (6)$$

$$\hat{\mathcal{G}} = \mathcal{G} \odot \sigma\left(\mathcal{L} W_{l\to g}\right) \quad (7)$$

$$Y = Concat\left(\hat{\mathcal{L}}, \hat{\mathcal{G}}\right) W_O \quad (8)$$

where $\hat{\mathcal{L}}$ and $\hat{\mathcal{G}}$ represent the gate-modulated local and global features, respectively. The terms $W_{g\to l} \in \mathbb{R}^{C_{\text{g}}\times C_{\text{l}}}$ and $W_{l\to g} \in \mathbb{R}^{C_{\text{l}}\times C_{\text{g}}}$ are learnable linear projection matrices, and $W_O \in \mathbb{R}^{C\times C}$ is the final output projection matrix. This bidirectional modulation mechanism allows the model to adaptively allocate weights between local and global information based on the input content, thereby balancing detail capture with global consistency modeling in complex maritime backgrounds.

Finally, the $C^2$-VRWKV Block significantly enhances the perception of small-scale ship structures through its local–global decoupling and gated fusion mechanism, all while preserving the original linear complexity and parallelization advantages of RWKV. This results in more robust feature representations under complex background conditions.

### C. Cross-Frequency Fusion Module

To mitigate the inherent locality loss of the 1D-scanning RWKV architecture and effectively suppress maritime background noise, we introduce the CFFM, as shown in Fig. 3. Distinct from traditional Fourier-domain transforms, the CFFM approximates frequency-domain decoupling through spatial-domain high-pass refinement, which isolates sharp object boundaries from low-frequency environmental clutter. The module processes features through asymmetric paths and leverages high-frequency guided cross-attention for feature enhancement.

1) *Asymmetric Dual-Path Processing:* Given an input feature $X \in \mathbb{R}^{B\times C\times H\times W}$, we first split the channels asymmetrically into $X_1$ and $X_2$ with a ratio of 1:3. $X_1$ is processed through a deep path consisting of stacked $3 \times 3$ convolutions to learn complex semantic features,

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denoted as $F_1$. Simultaneously, $X_2$ passes through a shallow path composed of a single $1 \times 1$ convolution to preserve raw structural information, resulting in $F_2$. This design provides feature sources with different levels of abstraction for the subsequent fusion.

2) *High-Frequency Detail Distillation:* To isolate target-related sharp edges from uniform sea backgrounds, we explicitly extract the high-frequency components, $F^{\uparrow}$, by subtracting the local intensity average from the original features

$$F^{\uparrow} = F - AvgPool_{k\times k}(F) \tag{9}$$

where $k$ denotes the pooling kernel size, which is set to 5 to ensure a robust estimation of local background intensity. This operation functions as a spatial high-pass filter that suppresses low-frequency environmental noise, thereby enhancing the discriminative saliency of high-frequency ship contours.

3) *Context-Aware Cross-Attention:* To prevent the amplification of stochastic background noise, we employ a mutual cross-attention strategy. Unlike standard attention modules, our channel attention (*ChannelAtt*) incorporates a $3 \times 3$ depthwise convolution (*DWConv*) to encode local context before pooling, while the spatial attention (*SpatialAtt*) utilizes a $1 \times 1$ convolution to distill interchannel saliency. The enhanced features $\hat{F}_1$ and $\hat{F}_2$ are defined as

$$\hat{F}1 = F_1 \odot \sigma\left(BN\left(Conv1 \times 1\left(F_2^{\uparrow}\right)\right)\right) \tag{10}$$

$$\hat{F}2 = F_2 \odot \sigma\left(AvgPool\left(DWConv3 \times 3\left(F_1^{\uparrow}\right)\right)\right) \tag{11}$$

where $\odot$ denotes element-wise multiplication and $\sigma$ represents the Sigmoid function. Through this cross-path gating, the semantics of the deep path guide the edge-extraction of the shallow path, ensuring that only high-frequency components consistent with ship structures are amplified.

4) *Adaptive Weighted Fusion:* Inspired by BiFPN, we perform an adaptive weighted fusion of the two enhanced features: a) $\hat{F}_1$ and b) $\hat{F}_2$, instead of a simple summation. We assign learnable weights $w_1$ and $w_2$ to the two branches, which are then normalized using a fast normalization scheme. The final fused feature is computed as

$$F_{\text{fused}} = \frac{w_1}{\sum_j w_j + \epsilon}\hat{F}_1 + \frac{w_2}{\sum_j w_j + \epsilon}\hat{F}_2 \tag{12}$$

where $\epsilon$ is a small constant for numerical stability and the weights $w_i$ are kept nonnegative. This data-driven fusion allows the CFFM to intelligently combine the strengths of both paths. Finally, the fused feature undergoes a final $1 \times 1$ convolution for information refinement. Through this design, the CFFM empirically learns to enhance the discrimination of targets against complex backgrounds.

### D. DS-MFA

To effectively address the challenge of multiscale variation in ship detection, we introduce the DS-MFA module. The core idea is to employ a prior-guided foreground–background separation mechanism, allowing for specialized processing of target and context regions. As shown in Fig. 4, the process begins with dual-stream separation. Given an input feature map $F \in \mathbb{R}^{B\times C\times H\times W}$ and a coarse prediction map $P_{\text{coarse}} \in \mathbb{R}^{B\times 1\times h\times w}$, we first use $P_{\text{coarse}}$ as a guidance prior. A foreground probability mask $M_{fg}$ is generated by upsampling the coarse map and applying a Sigmoid function. The input feature is then separated into a foreground stream $F_{fg}$ and a background stream $F_{bg}$

$$M_{fg} = \sigma\left(Upsample\left(P_{\text{coarse}}\right)\right) \tag{13}$$

$$F_{fg} = F \odot M_{fg} \tag{14}$$

$$F_{bg} = F \odot \left(1 - M_{fg}\right) \tag{15}$$

where $F_{fg}$ primarily contains features of potential ship targets of various scales, while $F_{bg}$ mainly consists of background context. $\sigma(\cdot)$ denotes the Sigmoid function and $\odot$ is element-wise multiplication.

Following the separation, the streams $F_{fg}$ and $F_{bg}$ undergo shared multifrequency stream enhancement. Both are processed by a shared stream processor inspired by SMFA, which internally splits each stream into a global context path and a local detail path. The global path computes an attention map

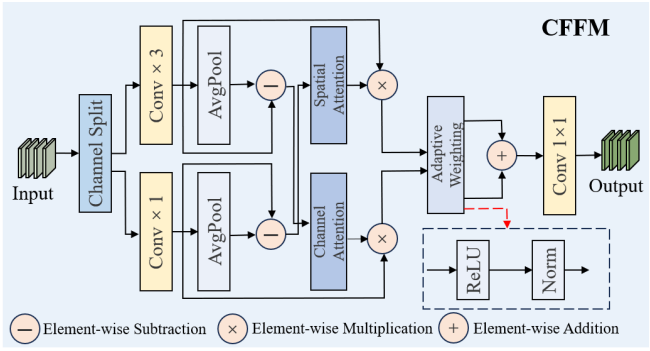



Fig. 3. CFFM module splits features into two streams and utilizes their high-frequency components to drive a cross-attention mechanism, allowing each stream to refine the other. The resulting features are intelligently fused through an adaptive weighting strategy, effectively suppressing background noise.

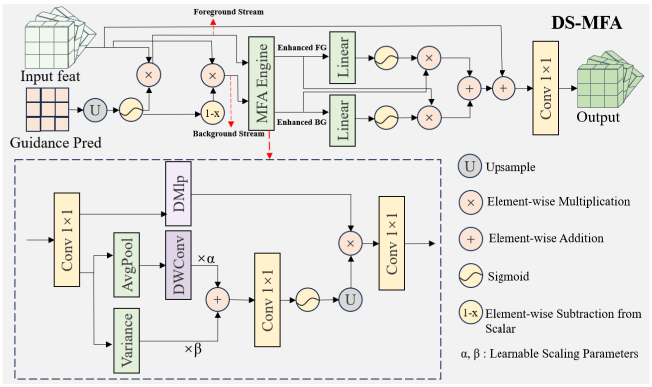



Fig. 4. DS-MFA module decouples features into prediction-guided foreground and background streams, which are enhanced by a shared MFA engine. These streams then interactively refine each other through mutual gating and are residually fused with the original input to enhance multiscale object features.

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using adaptive max-pooling and spatial variance, while the local path uses a lightweight depth-wise MLP to extract fine-grained details. The outputs of these two paths are fused to produce the enhanced streams $\hat{F}_{fg}$ and $\hat{F}_{bg}$.

Instead of a simple merge, we then introduce an interactive gating mechanism where the two enhanced streams mutually regulate each other. Two independent $1 \times 1$ convolutions, $\text{Conv}_{fg}$ and $\text{Conv}_{bg}$, generate gating signals from the opposing streams

$$F'_{fg} = \hat{F}_{fg} \odot \sigma\left(Conv_{bg}\left(\hat{F}_{bg}\right)\right) \tag{16}$$

$$F'_{bg} = \hat{F}_{bg} \odot \sigma\left(Conv_{fg}\left(\hat{F}_{fg}\right)\right). \tag{17}$$

In these equations, $\hat{F}_{fg}$ and $\hat{F}_{bg}$ are the enhanced foreground and background streams, respectively. This design allows contextual information from the background to suppress noise in the foreground, while foreground saliency helps the background stream to better identify and ignore target regions.

Finally, in the refinement stage, the interactively modulated streams $F'_{fg}$ and $F'_{bg}$ are integrated with the original input feature $F$ through a residual connection. A final $1 \times 1$ convolution, $\text{Conv}_{\text{out}}$, produces the module's output $F_{\text{out}}$

$$F_{\text{out}} = Conv_{\text{out}}\left(F + F'_{fg} + F'_{bg}\right). \tag{18}$$

Through this unified architecture, DS-MFA achieves a robust and efficient solution for handling significant scale variations in ship detection.

## IV. Experiments

### A. Datasets

*1) Airbus-Ship:*[1]*:* The Airbus-Ship dataset used in this study is a large-scale collection of optical remote sensing images with a total size of 29 GB, where each image has a resolution of $768 \times 768$ pixels. From this dataset, we selected 1000 valid samples containing ships of various sizes. These samples were divided into a training set (700 images), a validation set (200 images), and a test set (100 images) following a 7:2:1 split ratio.

*2) LEVIR-Ship [48]:* The LEVIR-Ship dataset comprises 3896 optical remote sensing images ($512 \times 512$ pixels) from China's GaoFen-1/6 satellites, featuring a 16-m spatial resolution. Following a rigorous quality screening, we selected 1900 high-quality samples of small-scale ships situated in complex background environments. These samples were subsequently partitioned into training (70%), validation (20%), and test (10%) sets at a 7:2:1 ratio.

The Airbus-Ship dataset is characterized by its significant multiscale properties, featuring ship targets that span a wide spectrum of sizes. A quantitative analysis, illustrated in Fig. 5(a), confirms this diversity. Despite the broad scale range, our investigation reveals that objects classified as small (area $<32 \times 32$ pixels) still constitute a substantial portion, accounting for 30.18% of the entire dataset. Crucially, the challenge of detecting these small instances is intensified during evaluation, as their proportion escalates dramatically to a dominant 76.74% within the designated test set [Fig. 5(c)].

[1]https://www.kaggle.com/c/airbus-ship-detection

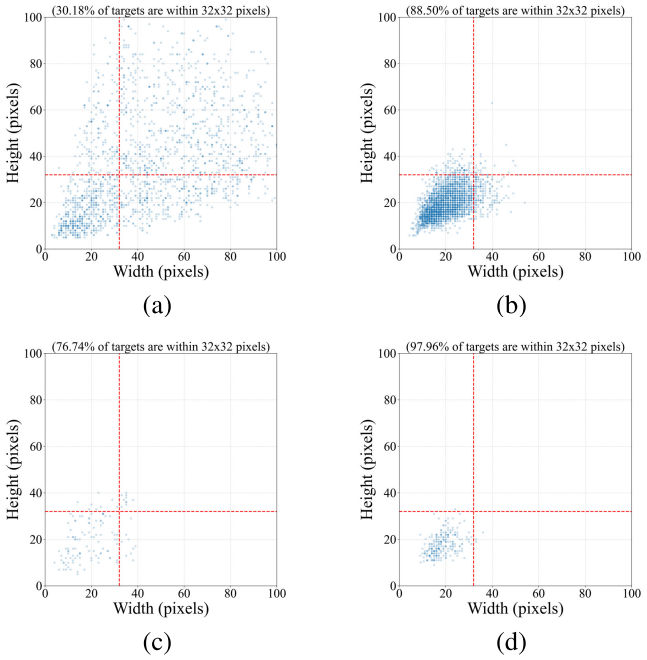



Fig. 5. Comparative analysis of ship target scales within two distinct datasets. We adopt the COCO benchmark's definition for small objects, classifying targets with dimensions under $32 \times 32$ pixels, a threshold indicated by the red dashed line. Specifically, (a) scale distribution for the complete Airbus-Ship dataset, with (c) same for its corresponding test split. Similarly, (b) and (d) target size distributions for the entirety of the LEVIR-Ship dataset and its respective test set.

This highlights the dataset's dual nature as both a multiscale and a small-target-focused benchmark. In contrast, the LEVIR-Ship dataset serves as a quintessential benchmark for small object detection, with its targets being inherently and overwhelmingly concentrated at the lower end of the size spectrum. Our statistical analysis [Fig. 5(b)] corroborates this, indicating that a remarkable 88.50% of all ship instances fall into the small object category. This extreme focus is further magnified in the test set, where small targets constitute an exceptionally high 97.96% of the samples [Fig. 5(d)]. The near ubiquity of small targets makes this dataset an ideal and rigorous testbed for evaluating a model's capabilities in handling diminutive objects.

### B. Implementation Details

*1) Experimental Settings:* All experiments were conducted on a workstation equipped with a single NVIDIA GeForce RTX 4090 GPU (24 GB), utilizing the PyTorch v1.8.0 framework running on Python v3.8.0. For optimization, we employed the stochastic gradient descent (SGD) optimizer with a momentum of 0.937 and a weight decay of $5e^{-4}$. The model was trained for 500 epochs using a batch size of 4, and the learning rate was initialized at 0.01. To address the extreme scale imbalance in the LEVIR-Ship dataset, where small objects represent 88.50% and 97.96% of the training and test sets, respectively, we adopted two targeted training strategies. First, we implemented a restricted geometric data augmentation approach by strictly constraining the minimum scaling factor. This ensures that minute targets maintain their structural integrity and avoid subpixel erosion during transformations.

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TABLE I
PERFORMANCE COMPARISON WITH SOTA METHODS ACROSS TWO BENCHMARK DATASETS

| Backbone | Venue | Params ↓ | GFLOPs ↓ | Size ↓ | LEVIR-Ship mAP(%) | Airbus-Ship mAP(%) | FPS ↑ |
|---|---|---|---|---|---|---|---|
| MobileNetV3-Small [38] | ICCV | 3.54 M | **6.3** | 7.2 MB | 61.15 | 68.61 | 61 |
| PP-LCNet [39] | CVPR | 3.74 M | 8.1 | 7.4 MB | 60.68 | 70.87 | 72 |
| ShufffeNetV2 [40] | ECCV | 3.78 M | 7.7 | 7.5 MB | 59.90 | 68.86 | 59 |
| MobileNetV3-Large [38] | ICCV | 5.20 M | 10.3 | 10.2 MB | 71.46 | 70.52 | 51 |
| GhostNet [41] | CVPR | 5.20 M | 8.4 | 10.4 MB | 65.58 | 71.94 | 53 |
| Efffcient-Lite0 [42] | ICML | 5.71 M | 11.5 | 11.2 MB | 66.40 | 70.08 | 61 |
| **Methods** | **Venue** | **Params ↓** | **GFLOPs ↓** | **Size ↓** | **LEVIR-Ship mAP (%)** | **Airbus-Ship mAP (%)** | **FPS ↑** |
| YOLOv5s | - | 7.05 M | 16.3 | 13.7 MB | 67.64 | 74.25 | 87 |
| YOLOv3-tiny [43] | CVPR | 8.67 M | 12.9 | 16.6 MB | 66.09 | 73.28 | 120 |
| YOLOv11s | - | 9.43 M | 21.5 | 54.5 MB | 68.40 | 76.01 | **157** |
| YOLOv12s [44] | CVPR | 9.30 M | 19.6 | 52.8 MB | 68.12 | 76.53 | 93 |
| YOLOv13s [45] | CVPR | 9.02 M | 21.0 | 52.6 MB | 68.45 | 76.70 | 107 |
| OWRT-DETR [46] | TGRS | 32.81 M | 108.0 | 125.9 MB | 70.75 | 77.91 | 106 |
| HSI-ShipDetectionNet [8] | TGRS | 4.15 M | 10.0 | 9.2 MB | 72.37 | 74.35 | 53 |
| ShipDetectionNet [47] | JSTARS | 6.05 M | 15.7 | 12.0 MB | 71.82 | 74.26 | 58 |
| VRWKV-T [20] | ICLR | 8.4 M | 67.9 | 16.3 MB | 65.52 | 74.84 | 101 |
| **Ours** | - | **0.53 M** | 8.6 | **2.5 MB** | **74.28** | **78.23** | 98 |

Second, we employed a multiscale loss rebalancing strategy by strategically increasing the classification and regression loss weights assigned to the high-resolution P3 detection head. This adjustment prevents the gradients of sparse targets from being overwhelmed by the dominant maritime background, ensuring sufficient supervision for microscale ship signatures throughout the optimization process. For the final evaluation, the model checkpoint that achieved the best performance on the validation set was selected.

*2) Comparison Methods:* To comprehensively evaluate the effectiveness of our proposed method, we conducted comparative experiments against 15 SOTA approaches. These include representative lightweight network architectures such as MobileNetV3-Small [38], PP-LCNet [39], ShuffleNetV2 [40], MobileNetV3-Large [38], GhostNet [41], and EfficientNet-Lite0 [42]; lightweight detection models from the YOLO family, including YOLOv3-tiny [43], YOLOv5s, and the more recent YOLOv11s, YOLOv12s [44], and YOLOv13s [45]; a Transformer-based detector, OWRT-DETR [46]; as well as lightweight models specifically designed for ship detection, such as HSI-ShipDetectionNet [8] and ShipDetectionNet [8]. Additionally, we include VRWKV [20] to further assess its adaptability in the ship detection context.

*3) Evaluation Metrics:* We employ five evaluation metrics to assess the performance of our model, including mean average precision (mAP), parameter count (Params), computational cost (GFLOPs), frames per second (FPS), and model size, to jointly evaluate the accuracy and efficiency of the proposed approach.

### C. Comparison With SOTA Methods

As listed in Table I, compared with other SOTA methods, our proposed $C^2F$-VRWKV achieves superior performance across both benchmark datasets while maintaining a high degree of efficiency. Specifically, on the LEVIR-Ship dataset, which is characterized by a high prevalence of small ships, our method outperforms all other SOTA models in terms of mAP, exceeding the next best method (HSI-ShipDetectionNet) by a margin of 2.64%. This improvement is attributed to the contextual clustering mechanism of the $C^2F$-VRWKV backbone, which effectively aggregates fine-grained morphological features to enhance the representation capability for small-scale targets.

On the Airbus-Ship dataset, which contains ships of multiple scales under complex background conditions, $C^2F$-VRWKV achieves the highest detection accuracy, outperforming YOLOv13s by 1.99% in mAP. These results demonstrate the effectiveness of the proposed DS-MFA and CFFM modules in modeling multiscale structural information while suppressing irrelevant environmental noise. Notably, our model exhibits a clear advantage in terms of the accuracy–complexity tradeoff. Compared with the recently proposed YOLOv13s, $C^2F$-VRWKV attains higher detection accuracy while requiring significantly fewer parameters. Furthermore, when compared with the Transformer-based OWRT-DETR, which employs 32.81M parameters and incurs 108.0 GFLOPs of computational cost, our method still maintains a 0.41% mAP advantage on the Airbus-Ship dataset, despite achieving a 92.04% reduction in computational complexity. Moreover, relative to the standard VRWKV-T baseline, the proposed enhancements result in a substantial reduction in parameters from 8.4M to 0.53M, while simultaneously improving the mAP on the LEVIR-Ship dataset by 13.37%. This observation validates that the proposed clustering-aware mechanism and frequency fusion strategy effectively compensate for the locality loss inherent in linear scanning architectures. Finally, with a compact model size of

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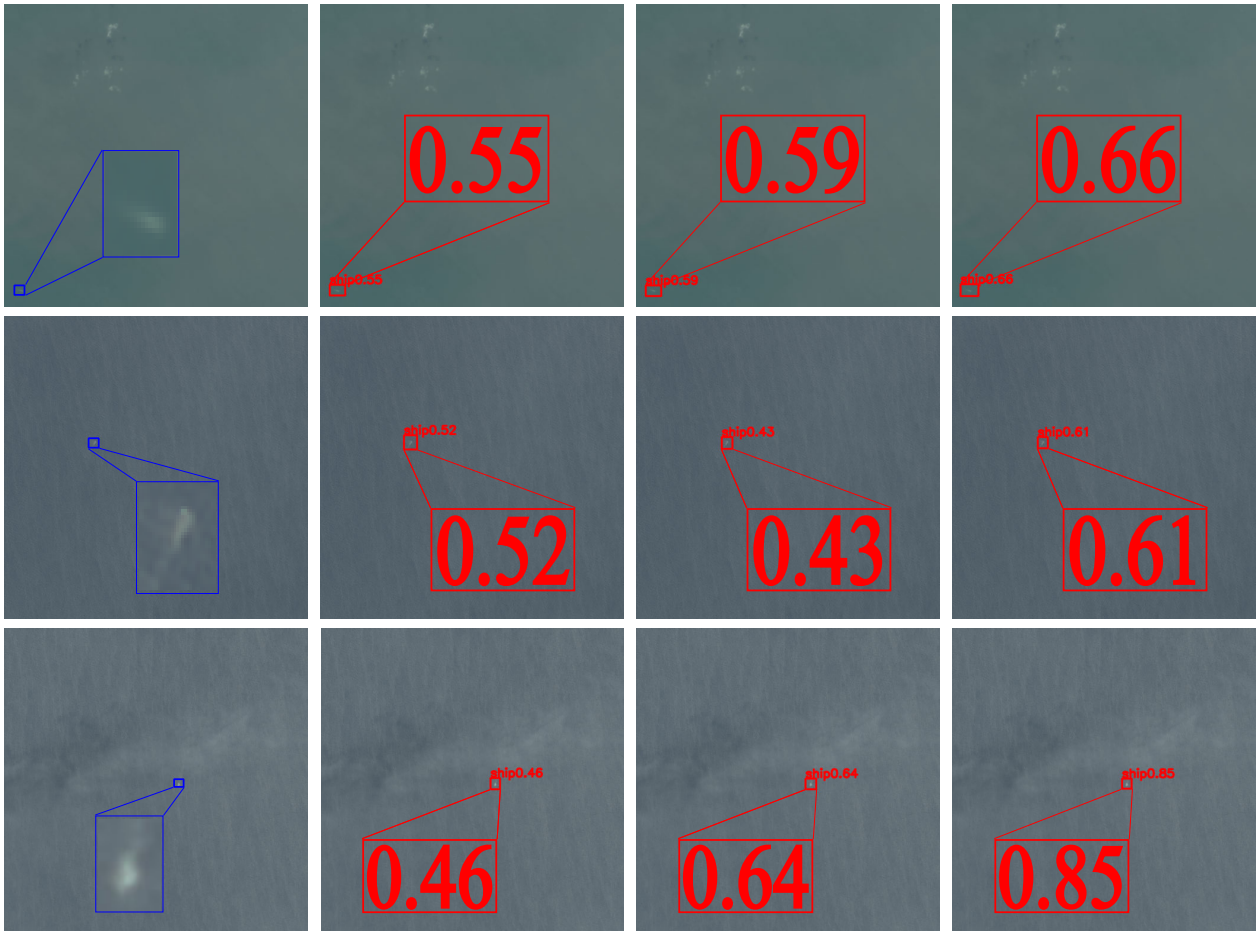



Fig. 6. Visualization comparison between our proposed method and other methods in small-scale ship detection scenarios.

2.5 MB and an inference speed of 98 frames/s, $C^2F$-VRWKV constitutes a practical and efficient solution for deployment on resource-constrained embedded platforms, making it particularly suitable for real-time maritime surveillance applications.

### *D. Visualization Results in Various Scenarios*

To assess the operational reliability and robustness of the proposed model in practical environments, qualitative comparative evaluations were conducted across several challenging maritime scenarios. The model's effectiveness in detecting small target ships is first verified in Fig. 6, which illustrates results under low-visibility conditions across three representative rows. In these scenes, ship targets are partially veiled by thin cloud cover or dense haze, leading to severe spectral attenuation and contrast degradation. While comparison models like YOLOv13s and baseline VRWKV suffer from significantly lower confidence scores, our proposed model maintains high-confidence localization. This confirms that the $C^2F$-VRWKV backbone successfully aggregates fine-grained morphological features, significantly enhancing the representation capability for small-scale ships.

The model's discriminative power under various environmental disturbances is first investigated in Fig. 7 across five representative rows. In scenarios with heavy cloud cover (Rows 1 and 2), the bright cloud edges often act as "pseudo-targets" that mislead conventional detectors. Our proposed model successfully suppresses this background clutter and maintains focus on true ship signatures. For scenes featuring dense sea ripples (Row 3), the model effectively distinguishes ship hulls from high-frequency periodic wave textures. In cases of intense sun glint (Row 4), the numerous bright reflection points often trigger false alarms in other methods, whereas our method achieves a superior confidence score, effectively filtering out specular noise. Furthermore, when tested in large-scale coastal scenarios containing dense terrestrial infrastructures (Row 5), our model precisely identifies ship targets while maintaining a zero false-alarm rate on land-based structures. This superior performance confirms the robustness and reliability of our method across diverse and cluttered real-world maritime backgrounds. Finally, the model's sensitivity to dramatic scale variations is further examined in Fig. 8, where a comparative analysis is performed specifically against the baseline VRWKV. These scenarios feature the coexistence of multiscale ship targets, ranging from large cargo ships to minute fishing boats. The qualitative results demonstrate that the standard VRWKV architecture often fails to capture multiscale dependencies, resulting in missed detections of smaller ships. In contrast, our proposed method accurately localizes

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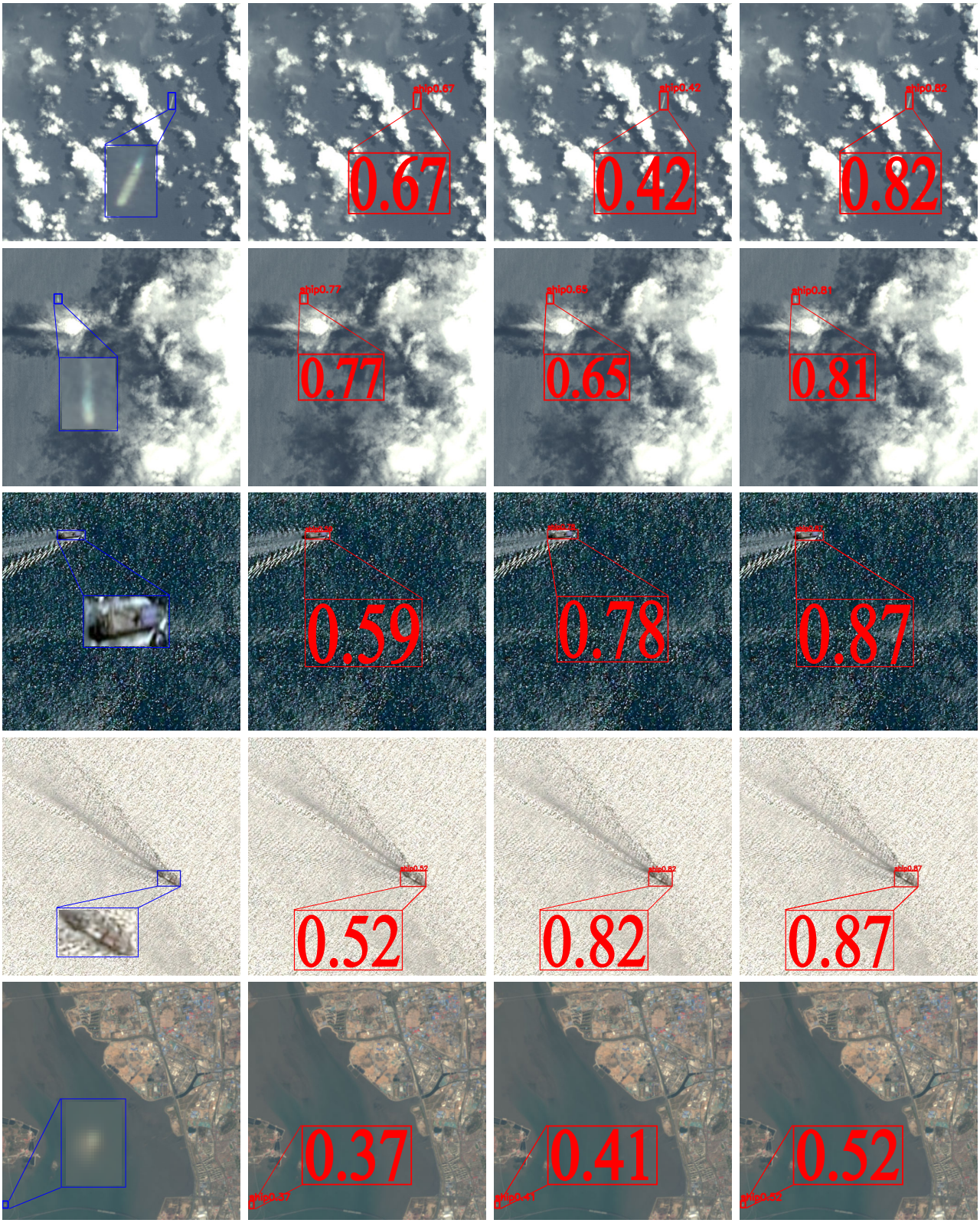



Fig. 7. Visual comparison in various complex maritime scenarios.

all ship targets regardless of their scale. This validates the effectiveness of the DS-MFA module in modeling diverse ship structures and ensuring scale-invariant feature representation even in high-complexity environments.

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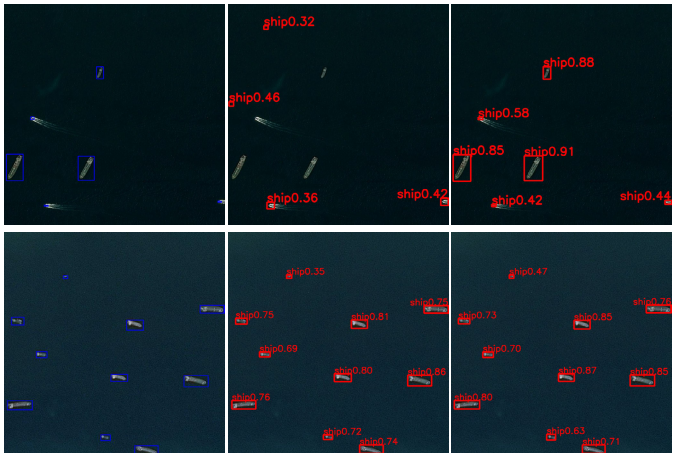



Fig. 8. Visualization comparison between our proposed method and the baseline VRWKV in multiscale scenarios.

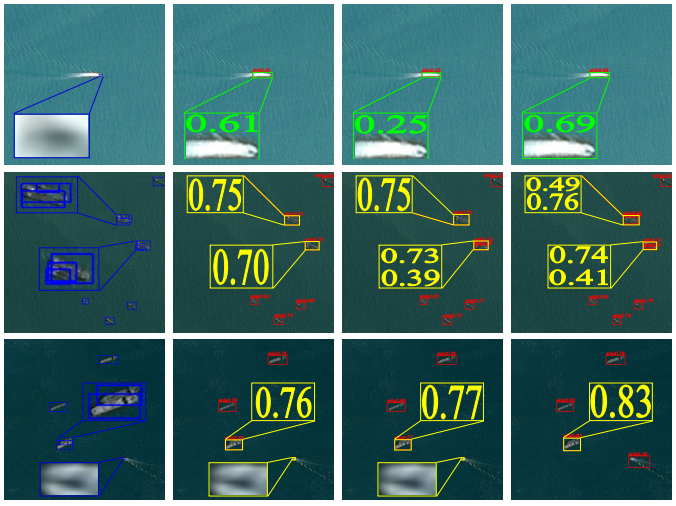



Fig. 9. Representative failure cases and boundary limitations of the proposed $C^2F$-VRWKV model. The yellow annotations represent missed detections, while the green annotations indicate false positives.

### E. Error Analysis and Limitations

While the proposed $C^2F$-VRWKV achieves state-of-the-art performance, a comparative examination of its failure cases against baseline and contemporary models is essential for understanding the physical boundary conditions of the architecture. As shown in Fig. 9, we identified two primary modes of failure common to maritime detection: false positives induced by dynamic wakes and missed detections in ultradense clusters, which stem from the inherent tradeoffs in our structural designs.

First, regarding the top row of Fig. 9, all compared models exhibit false positives where bright, dynamic ship wakes are misidentified as actual hulls. This phenomenon is a direct consequence of the high-frequency sensitivity inherent in modules such as our CFFM. Since foaming wakes generate intense spatial gradients similar to the structural signatures of ship hulls, the spatial high-pass filtering process occasionally amplifies these disturbances. Notably, however, our $C^2F$-VRWKV maintains higher confidence scores on true targets compared to YOLOv13s and the baseline VRWKV, suggesting a more reliable discrimination between actual vessels and environmental noise even when both are misdetected.

Second, the bottom two rows illustrate missed detections in ultradense ship clusters where targets are moored shoulder-to-shoulder. This common challenge is primarily linked to the local window clustering mechanism within our $C^2$-VRWKV backbone. While this mechanism enhances local feature aggregation, the extreme spatial proximity of targets causes their feature representations to heavily entangle during the window-based processing phase. Such contextual blending blurs discriminative boundaries, making it difficult for the subsequent suppression stages to isolate highly overlapping instances. Nevertheless, as evidenced in the second row, our model still captures a higher number of valid targets in these dense scenes compared to the baselines. Furthermore, for the sailing vessel in the third row, our architecture successfully generates a detection while both YOLOv13s and the baseline VRWKV fail completely, further demonstrating the superior sensitivity of our refined feature representations.

These identified failure cases chart clear pathways for future optimization. We plan to explore multitemporal or kinematic feature integration to distinguish the static hull from the dynamic wake based on temporal consistency. To resolve the feature entanglement issue inherent to the $C^2$-VRWKV backbone, future work will investigate oriented bounding box (OBB) representations to better separate tightly packed instances.

TABLE II
ABLATION STUDY ON THE AIRBUS-SHIP DATASET

| $C^2$-SpatialMix | CFFM | DS-MFA | Params ↓ | GFLOPs ↓ | Size ↓ | mAP (%) |
|---|---|---|---|---|---|---|
| ✗ | ✗ | ✗ | **0.43 M** | **5.9** | **2.1 MB** | 71.51 |
| ✓ | ✗ | ✗ | 0.43 M | 5.9 | 2.1 MB | 74.83 |
| ✗ | ✓ | ✗ | 0.46 M | 6.8 | 2.2 MB | 76.72 |
| ✗ | ✗ | ✓ | 0.49 M | 7.7 | 2.4 MB | 76.92 |
| ✓ | ✓ | ✗ | 0.46 M | 6.8 | 2.2 MB | 76.60 |
| ✓ | ✗ | ✓ | 0.49 M | 7.7 | 2.4 MB | 77.20 |
| ✗ | ✓ | ✓ | 0.53 M | 8.6 | 2.5 MB | 77.37 |
| ✓ | ✓ | ✓ | 0.53 M | 8.6 | 2.5 MB | **78.23** |

### F. Ablation Studies

1) Table II reports the ablation study results on the Airbus-Ship dataset to evaluate the individual and collective contributions of the proposed components. To ensure structural integrity when specific modules are ablated, we establish strictly controlled alternative operations: When the $C^2$-SpatialMix is ablated, we revert the backbone block to the original VRWKV architecture, utilizing a pure 1-D global linear scanning mechanism without local clustering. When the CFFM is ablated, we replace its frequency-domain decoupling with a standard top-down feature fusion paradigm, employing simple concatenation followed by a $1 \times 1$ convolution for channel alignment. When the DS-MFA is ablated, we bypass the prior-guided separation and replace the entire refinement path with a standard convolutional

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block before the detection head. Starting from a baseline configuration with an mAP of 71.51%, the step-wise integration of $C^2$-SpatialMix, CFFM, and DS-MFA consistently enhances detection performance, culminating in an optimal mAP of 78.23%. Specifically, the integration of the $C^2$-SpatialMix module provided a significant performance boost, improving the mAP by 4.64% compared to the baseline. It is noteworthy that the associated increase in parameter count and computational complexity remained negligible, as shown by the rounded metrics in Table II. This result underscores the high efficiency of the contextual clustering mechanism in optimizing the 1-D scanning paradigm, allowing the model to capture fine-grained spatial dependencies with nearly constant computational overhead.

Furthermore, the CFFM and DS-MFA modules offer a highly efficient tradeoff between precision gains and computational footprint. The inclusion of CFFM alone improved the detection accuracy by 7.29%, while only incurring a marginal increase of 0.03M parameters and 0.9 GFLOPs. This confirms that the spatial high-pass refinement is an effective strategy for suppressing maritime clutter. Similarly, employing DS-MFA as a standalone enhancement increased the mAP by 7.57%, demonstrating the effectiveness of prior-guided foreground–background separation in handling multi-scale variations. When these components are combined, a clear synergistic effect is observed, leading to a peak performance of 78.23%. Notably, even in its complete configuration, $C^2F$-VRWKV maintains an extremely compact profile of 0.53M parameters and 8.6 GFLOPs. This represents a total absolute precision improvement of 9.40% over the baseline with an overall parameter increase of only 0.1M. Such quantitative evidence confirms that the proposed architecture achieves a superior balance between structural efficiency and robust ship detection in complex environments.

2) *Ablation of the Bidirectional Gated Fusion Mechanism:* To provide a more in-depth analysis of exactly how the bidirectional gated fusion mechanism within the $C^2$-SpatialMix module enhances the model's representational capability, we design a specific microablation study accompanied by qualitative feature visualizations. The core objective of this experiment is to validate the necessity of the gating mechanism and to intuitively reveal how local fine-grained structures and global macrosemantics mutually calibrate each other to extract pure target signals from complex maritime clutter.

We first quantitatively isolate the fusion logic by evaluating four variants while keeping the rest of the network intact (including CFFM and DS-MFA). As shown in Table III, merely fusing the local ($L$) and global ($G$) features via standard concatenation (Simple Concat) yields a significantly suboptimal mAP of 75.33%. This drastic degradation occurs because standard concatenation indiscriminately propagates both the high-frequency wave noise extracted by the local branch and the spatially diffuse boundaries from the global branch to subsequent layers. Although unidirectional gating strategies provide partial information filtering, which improves the mAP to 77.76% for unidirectional ($L \rightarrow G$) and 77.89% for unidirectional ($G \rightarrow L$), they still fall short of optimal feature representation. In contrast, our proposed Bidirectional gating dynamically cross-calibrates both branches, achieving the peak accuracy of 78.23%. This empirical evidence confirms that simultaneous cross-modulation is mathematically critical for maximizing discriminative representation without introducing additional computational overhead.

To intuitively reveal the physical process behind this representational enhancement, we extract and visualize the intermediate feature maps from within the $C^2$-SpatialMix module during inference on a cluttered maritime image, as shown in Fig. 10. Initially, the raw local feature ($L$) in Fig. 10(b) suffers from severe "feature drowning," as it is heavily corrupted by high-frequency sea wave noise which shares similar textural properties with the target. However, guided by its macrosemantic understanding, the global branch

TABLE III
ABLATION OF FUSION STRATEGIES FOR REPRESENTATIONAL ENHANCEMENT IN $C^2$-SPATIALMIX

| Fusion Strategy | Formula | mAP (%) |
|---|---|---|
| Simple Concat | $Y = \text{Concat}(L, G)W_O$ | 75.33 |
| Unidirectional ($L \rightarrow G$) | $Y = \text{Concat}(L, \hat{G})W_O$ | 77.76 |
| Unidirectional ($G \rightarrow L$) | $Y = \text{Concat}(\hat{L}, G)W_O$ | 77.89 |
| Bidirectional | $Y = \text{Concat}(\hat{L}, \hat{G})W_O$ | **78.23** |

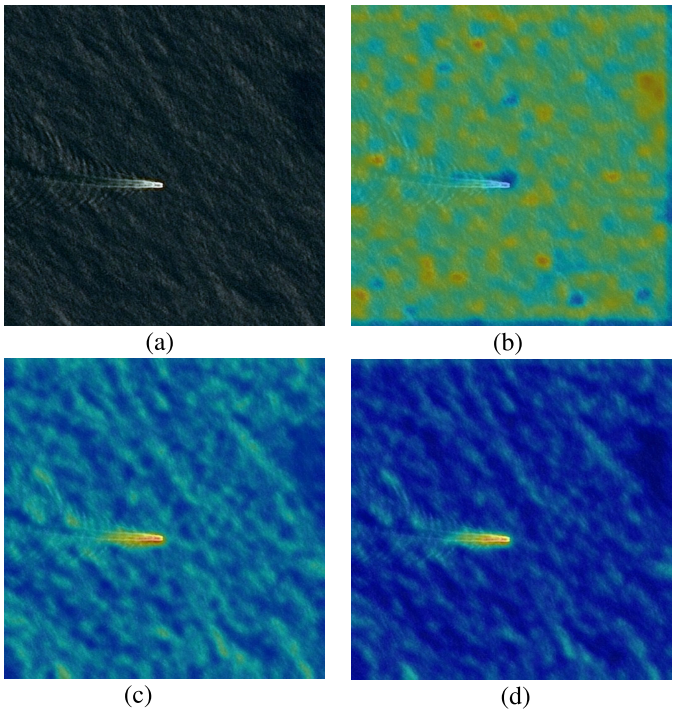



Fig. 10. Visualization of the feature purification process by the bidirectional gated fusion mechanism. (a) Input image. (b) Local feature ($L$). (c) Global gate [$\sigma(\mathcal{G}W_{g\rightarrow l})$]. (d) Purified feature ($\hat{L}$).

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TABLE IV
ABLATION STUDY ON THE DEPTH DISTRIBUTION STRATEGY OF STACKED BLOCKS ACROSS THE THREE STAGES OF THE $C^2$-VRWKV BACKBONE

| $\{d_1, d_2, d_3\}$ | **Params** ↓ | **GFLOPs** ↓ | **Size** ↓ | **mAP (%)** |
|---|---|---|---|---|
| $\{1, 1, 1\}$ | **0.4716 M** | **8.4** | **2.3 MB** | 75.68 |
| $\{1, 2, 1\}$ | 0.5005 M | 8.8 | 2.4 MB | 75.91 |
| $\{1, 2, 2\}$ | 0.5544 M | 8.9 | 2.7 MB | 77.07 |
| $\{2, 1, 1\}$ | 0.4844 M | 9.1 | 2.4 MB | 77.12 |
| $\{2, 1, 2\}$ | 0.5384 M | 9.2 | 2.7 MB | 78.16 |
| $\{2, 2, 1\}$ | 0.5133 M | 9.4 | 2.5 MB | 77.92 |
| $\{2, 2, 2\}$ | 0.5672 M | 9.6 | 2.7 MB | 77.01 |
| $\{1, 1, 2\}$ | 0.5256 M | 8.6 | 2.5 MB | **78.23** |

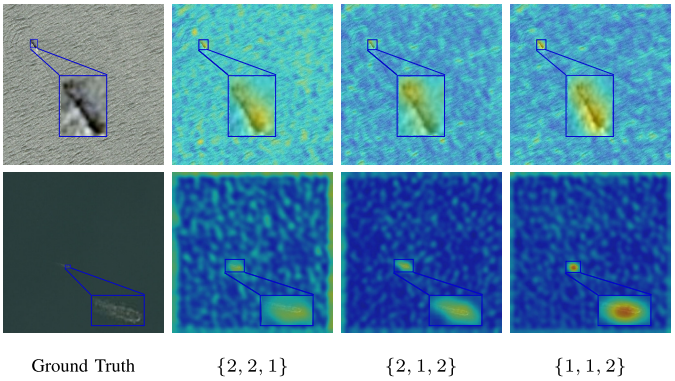



Fig. 11. Visualization comparison of different architecture configurations.

generates a highly precise spatial mask, $\sigma(G \rightarrow l)$ [Fig. 10(c)]. This mask intelligently assigns high activation weights exclusively to the true ship target while allocating near-zero weights to the surrounding wave clutter. Upon element-wise gating modulation, the pervasive background noise is decisively eliminated, yielding a highly purified and sharp representation of the small ship in Fig. 10(d). Symmetrically, the local branch generates a reciprocal mask to force the diffuse global attention to align with precise 2-D topological boundaries. By dynamically cross-calibrating in this bidirectional manner, the mechanism fundamentally resolves the conflict between noise suppression and detail preservation, which is exactly how it enhances the model's representational capability in complex remote sensing scenarios.

3) *Ablation on Backbone Depth Distribution:* To systematically investigate the optimal architectural design for ship detection, we perform an extensive ablation study on the depth distribution strategy $\{d_1, d_2, d_3\}$ across the three stages of the $C^2$-VRWKV backbone. In optical remote sensing imagery, ship targets are typically minute and often embedded within vast, high-frequency clutter such as sea waves or clouds. Consequently, the backbone must strike a balance between retaining fine-grained spatial details and extracting robust, noise-resistant global semantics.

   As shown in Table IV, we quantitatively compare various block allocation strategies. Shallow configurations, such as $\{1, 1, 1\}$, lack sufficient representational capacity, yielding a mAP of only 75.68%. Conversely, early-heavy architectures allocate excessive computational resources to high-resolution shallow layers. While preserving local textures, these designs fail to adequately suppress macroscale background interference, resulting in limited performance gains despite increased GFLOPs. Furthermore, uniformly scaling the network depth significantly inflates the parameter count to 0.5672M and increases GFLOPs to 9.6, yet the accuracy degrades to 77.01%, indicating severe feature smearing of tiny objects in deep layers. In contrast, our proposed late-heavy configuration $\{1, 1, 2\}$ strategically allocates a second block specifically to the deepest stage (Stage 3). By keeping Stages 1 and 2 lightweight, we efficiently extract necessary spatial details without wasting computational overhead on high-resolution noise. The additional block in Stage 3, where the global receptive field is maximized, acts as a crucial "semantic purifier." It leverages high-level macrocontext to decisively silence the background clutter, providing a highly confident semantic anchor for the subsequent FPN modules. This design achieves the peak accuracy of 78.23% mAP while maintaining an extremely lightweight profile of 0.5256M parameters and 8.6 GFLOPs, proving to be the optimal sweet spot for accuracy and efficiency.

   To intuitively demonstrate the representational superiority of this specific depth distribution, we visualize the backbone output feature maps across different configurations in Fig. 11. In the early-heavy $\{2, 2, 1\}$ design, the feature maps are notably "noisy;" the network struggles to distinguish the ship from high-frequency wave textures, resulting in widespread false-positive activations across the sea surface. The $\{2, 1, 2\}$ design exhibits improved background suppression, yet it suffers from weak and diffuse target activation, failing to sharply localize the ship hull. In striking contrast, our optimal $\{1, 1, 2\}$ configuration achieves a superior disentanglement of the target from the background. As observed in the detailed feature representations, the wave clutter is effectively suppressed, maintaining a low-response background, while the feature energy remains intensely focused on the ship target with a distinct, high-intensity peak. This visual evidence confirms that allocating more capacity to the deepest stage acts as an effective semantic purifier, enhancing the target's discriminative saliency against complex maritime environments.

4) *Ablation on the Foreground–Background Separation Strategy Within DS-MFA:* To specifically elaborate on the advantages of the foreground–background separation strategy within the DS-MFA module, we provide a comprehensive internal feature visualization in Fig. 12. This dual-stream strategy is explicitly designed to address the inherent challenge in remote sensing where the weak signatures of multiscale ships are easily diluted and overwhelmed by dominant maritime background clutter. The top row in Fig. 12 illustrates the mechanistic operation of this separation process. Given a

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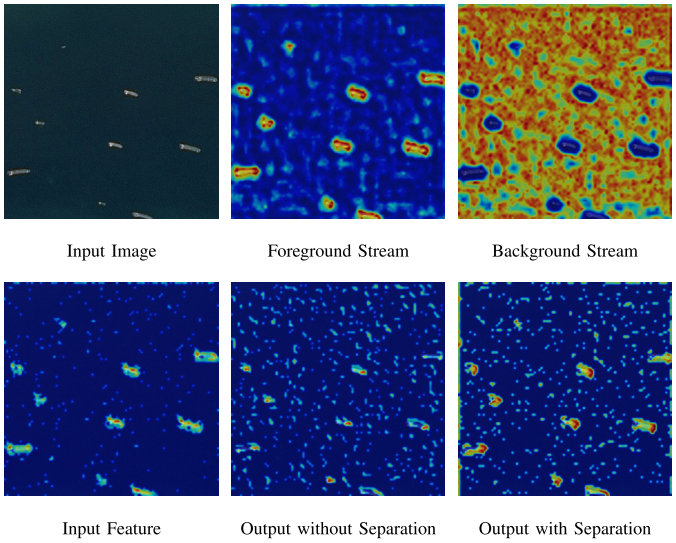



Fig. 12. Foreground–background separation mechanism and visualization comparison.

cluttered input image, the DS-MFA module leverages a coarse prediction before decoupling the features into two distinct streams. The foreground stream successfully isolates target-related pixels across various scales, representing them as highly focused, high-energy activations while effectively silencing the nontarget regions. Symmetrically, the background stream captures the rich environmental wave textures, leaving precise semantic voids exactly at the ship coordinates. This physical decoupling provides the subsequent shared MFA engine with two specialized, noninterfering statistical domains for refinement.

The bottom row substantiates the performance superiority of this strategy through a direct comparison against a nonseparated baseline. The input feature, derived from the preceding network layers, inherently contains an entangled mixture of target signals and complex wave noise. When processing this entangled feature, the single-stream baseline struggles to allocate attention efficiently; its enhancement of the ship targets remains relatively weak, and it fails to adequately suppress the surrounding high-frequency background clutter. In striking contrast, by employing our interactive separation and cross-calibration strategy, the separated output achieves a substantially improved signal-to-noise ratio. The background wave noise is decisively compressed into a lower-energy state, while the feature representation is intensely and sharply amplified specifically on the ship targets of varying sizes. This visual evidence unequivocally confirms that foreground–background separation is a physically indispensable mechanism to prevent information dilution, enabling specialized and high-contrast feature refinement for multiscale targets in complex remote sensing scenarios.

## V. Conclusion

In this article, we proposed $C^2F$-VRWKV, a lightweight and efficient network based on vision RWKV for high-precision multiscale ship detection. $C^2F$-VRWKV combines a $C^2$-VRWKV backbone with CFFM and DS-MFA fusion modules to enhance feature representation for ships of various scales against complex backgrounds, all with a minimal parameter count. Equipped with contextual clustering for small targets, CFFM for background suppression, and DS-MFA for multiscale modeling, the network excels at capturing diverse ship characteristics. Experiments on two public datasets show $C^2F$-VRWKV's exceptional performance, especially in challenging maritime scenarios. On the demanding Airbus-Ship dataset, it achieved a mAP of 78.23% with only 0.53M parameters, demonstrating its effectiveness for real-world ship detection and suitability for edge devices. Future work will enhance model robustness using diverse data while optimizing the architecture for greater accuracy and computational efficiency.

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